

Selected Topics in Visual Recognition using Deep Learning

HW2 Report

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My Github Code link: https://github.com/Pakomu/CV_DeepLearning_2026/tree/main

I. Introduction

For this assignment, inspired by the lessons mentioned and I tested the training speed, my core methodology used the deformable DETR according to its convergence speed. Also, I implemented the augmentation method by applying the albumentation package. However, I finally cannot reach the pass score of this homework.

II. Method

In the data processing part, I apply the Albumentation package to get a better and fast data augmentation method. These are the processing methods that I choose: including the Blur, adding noise, Color jitter(contrast, brightness, hue, saturation), CLAHE, which slice the original image into parts and do the different contrast process on each block. There is one important thing that cannot use the rotate or the clip process which would affect the image outcome and become bad data.

Then, I called the Deformable DETR from pytorch, here are my hyperparameters:

backbone	Resnet50
Batch size	2
accumulation	8
learning rate	1e-4
backbone learning rate	1e-5
weight decay	1e-4
num_labels	NUM_CLASSES=11
encoder_layers	2
decoder_layers	4
num_queries	100
auxiliary_loss	True

The encoder layer and the decoder layers are my experiment to find the best for this homework.

Then I train my epoch like the previous homework. There are some special for this homework:

1. The training time is long, and it is difficult to do other things during the training due to the GPU utilization. Therefore, I save all the parameters including checkpoint, learning rate, optimizer status etc. to make sure I can stop the training and can restart next time.

2. When the decoder, encoder layers are increased, my GPU card is overloaded for high batch size. So I decrease the batch size to 2. And I listen to the advice that when batch size is small, it is better to continue for some accumulation step and update. My final result is set a batch size = 2 and accumulation step=8
3. This homework detr(deformable detr) has split the network into two parts: CNN backbone and the Transformer decoder layers. And I set the different learning rate for two different parts. Because we have already applied the pretrained weight on CNN backbone(resnet50), I set the smaller learning rate.

Finally, I also compute the valid_loss to determine the best model. Also, I save every 5 epoch to ensure the later stage.

About the test part, Although DETR should not apply to the NMS, I think that I don't have enough or good training, so it cannot reduce my score on irrelevant parts. I still applying for the NMS.

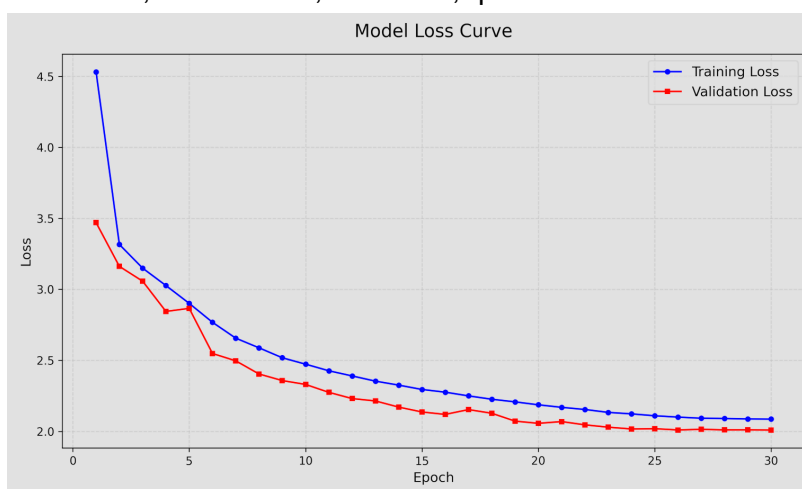
III. Additional experiments

My additional experiments are partially based on the training time because I didn't expect that the training of DETR would take so much time. At first I applied the original decoder , encoder layer and found out that it took lots of time because I can only use batch size = 1(about 2 hr per epoch). Then, I decrease both the encoder, decoder layer to 2. Also, after I observed the dataset in eyes. I didn't think it needed so many queries. Then I try to decrease the num_queries from 300 to 100 and test the training time. So far, I have a stable hyperparameter that can control the training time in 30 minutes per epoch. So I try to improve the layer, queries count slightly to get a better preformance. I found out the encoder layer approval affects more than the decoder layer. Therefore, I set the parameters that I mentioned in the Method part as my best hyperparameter for this homework.

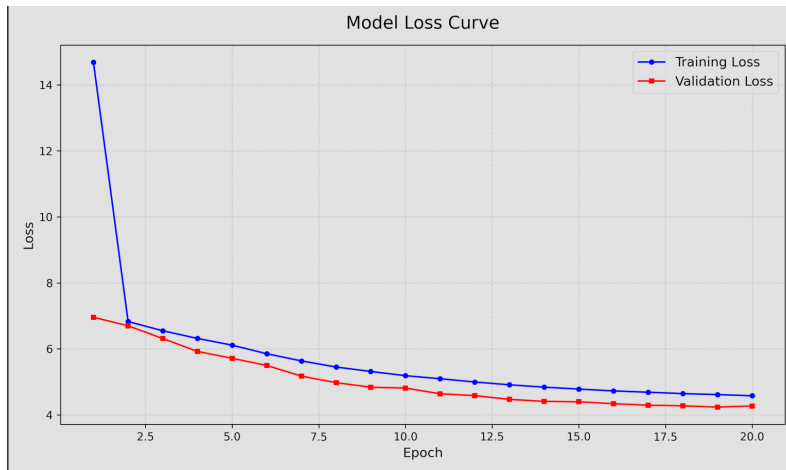
IV. Results

I record my loss value from each training epoch in different hyperparameters. The thing that captures my eyes is that the convergence times for both DETR and the deformable DETR don't reach as high as the lecture note mentioned, which is so low that I am not sure if it was my problem.

decoder=2, encoder = 2, batch = 4, queries = 100



decoder=2, encoder = 4, batch = 2, queries = 100



This is my result snapshots:

61	airozo	2026-04-18 11:42	685495	111550148	0.32
62	stanly	2026-04-20 10:57	687822	314553035	0.32
63	rasparc	2026-04-21 00:58	688836	313540041	0.31
64	313551003	2026-04-19 23:34	687326	313551003	0.3
65	olepal	2026-04-21 16:17	689782	111550202	0.3
66	titouan_gagneux	2026-04-14 16:47	678882	114550830	0.3
67	111550017	2026-04-21 16:31	689808	111550017	0.28
68	powei989	2026-04-20 12:20	687891	314553039	0.28
69	314554036	2026-04-21 09:11	689248	314554036	0.28
70	ywliu	2026-04-18 16:38	685720	313553049	0.28
71	412551033	2026-04-20 15:29	688105	412551033	0.24
72	pakomu	2026-04-21 20:43	690128	111550028	0.2
73	rawnjt	2026-04-21 20:01	690069	314540030	0.15
74	rusbono	2026-04-21 16:27	689799	314553801	0.08
75	xxxxxx123	2026-04-16 01:53	681492	test	0.07
76	piggypsychoduck	2026-04-21 16:44	689836	412540009	0.0
77	marksai	2026-04-21 16:14	689776	111550133	0.0
78	112550183	2026-04-15 19:39	680956	112550183	0.0
79	keliping	2026-04-21 10:52	689409	314515802	0.0
80	skayskay	2026-04-20 20:27	688476	109550202	0.0

V. Reference

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[2] Zhu, X., Su, W., Lu, L., Li, B., Wang, X., Dai, J.: Deformable DETR: Deformable Transformers for End-to-End Object Detection. In: International Conference on Learning Representations (ICLR). (2021).

[3] He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770--778 (2016).

[4] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Chintala, S., et al.: PyTorch: An Imperative Style, High-Performance Deep Learning Library. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 8024--8035 (2019).